

CHERRY POINT MCAS, NC, USA

WMO: 723090

Lat: 34.900N

Lon: 76.883W

Elev: 29

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13754

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.7	27.2	5.6	7.3	30.4	10.3	9.3	32.9	26.4	45.7	24.3	45.5	7.3	10	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	91.6	77.8	89.9	77.2	88.0	76.4	80.4	87.4	79.3	86.1	78.2	84.7	9.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.7	149.1	84.1	77.3	142.4	83.1	76.3	137.3	82.1	43.9	87.7	42.7	86.1	41.6	84.7	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.4	20.0	18.1	DB	16.3	96.4	4.2	2.5	13.3	98.2	10.9	99.7	8.5	101.1	5.5	103.0	(4)
(5)			WB	14.4	82.9	3.7	1.2	11.8	83.8	9.6	84.5	7.5	85.2	4.8	86.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	45.2	47.8	53.6	62.1	69.8	77.1	79.8	79.1	74.5	65.1	55.1	49.3		
	DBStd	14.26	10.24	9.17	8.65	7.35	6.28	4.63	3.37	3.37	4.48	7.45	8.29	9.41		
	HDD50	599	219	138	63	4	0	0	0	0	0	2	41	132		
	HDD65	2531	615	485	366	140	27	1	0	0	2	93	310	492		
	CDD50	5452	71	76	174	367	615	814	924	901	734	471	195	110		
	CDD65	1909	2	3	11	53	177	365	459	436	286	97	14	6		
	CDH74	15264	3	10	54	301	1261	3101	4275	3815	1830	552	52	10		
(13)	CDH80	4772	0	0	4	39	330	1051	1514	1295	449	88	2	0		
(14)	Wind	WSAvg	8.2	8.8	9.2	9.8	9.8	8.6	7.7	7.2	6.6	7.8	7.4	8.1		
(15) Precipitation	PrecAvg	54.6	4.2	3.9	4.0	3.1	5.0	5.0	7.0	7.4	5.2	3.2	3.1	3.6		
	PrecMax	68.2	9.5	9.3	8.8	9.3	12.2	11.3	14.6	19.2	13.7	10.8	9.2	9.2		
	PrecMin	41.9	0.6	0.6	0.9	0.1	2.2	1.0	1.3	2.1	0.2	0.5	0.4	0.4		
	PrecStd	7.2	1.9	2.0	1.8	2.0	2.7	2.4	3.0	4.2	3.4	2.4	2.1	2.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.4	75.2	79.4	84.2	91.2	94.3	94.4	93.5	89.7	86.1	79.2	75.4		
		MCWB	65.0	66.2	65.9	68.7	74.4	76.8	79.2	79.1	76.2	74.2	69.8	67.5		
	2%	DB	69.9	71.1	74.9	80.6	86.4	90.8	91.7	90.5	87.1	82.2	75.1	71.9		
		MCWB	63.9	62.2	63.6	67.3	72.7	76.5	78.6	78.1	74.7	72.2	67.2	65.6		
	5%	DB	66.3	67.3	71.8	77.5	83.4	88.4	89.6	88.6	84.6	79.6	72.4	69.2		
		MCWB	61.1	60.3	61.5	65.9	71.6	75.6	78.0	77.4	73.9	70.7	65.8	64.7		
	10%	DB	62.8	63.8	68.4	74.7	81.0	85.9	87.5	86.6	82.4	77.0	69.6	65.7		
		MCWB	58.1	57.6	60.6	64.5	70.1	74.7	77.1	76.3	73.5	69.6	63.6	62.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.0	68.7	68.7	73.2	77.4	80.6	82.3	82.1	78.6	77.6	73.1	70.4		
		MCDB	71.4	73.1	74.3	78.9	85.8	88.6	89.8	89.3	85.1	81.7	76.7	73.2		
	2%	WB	65.2	65.5	66.4	70.7	75.2	78.7	80.5	80.3	77.4	75.5	70.0	67.5		
		MCDB	68.4	69.3	71.6	76.5	82.4	86.8	87.8	86.9	83.4	79.7	73.1	70.0		
	5%	WB	62.3	62.4	64.4	68.5	73.5	77.5	79.5	79.2	76.4	73.3	67.2	65.0		
		MCDB	65.8	65.8	69.2	74.4	80.3	85.3	86.5	85.3	81.8	77.5	70.8	68.0		
	10%	WB	58.7	58.7	62.3	66.6	72.0	76.3	78.3	78.0	75.4	71.2	64.5	62.0		
		MCDB	62.3	63.4	67.3	72.7	78.6	83.2	84.8	84.0	80.3	75.4	68.4	65.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.3	19.8	20.3	20.0	18.4	16.4	15.0	14.9	14.7	18.1	19.7	19.0		
		MCDBR	23.4	23.5	23.3	21.5	20.3	18.6	17.2	17.0	16.8	19.2	21.4	21.1		
		MCWBR	18.0	16.6	13.9	11.8	9.6	7.8	7.6	7.5	7.4	10.4	14.6	16.0		
	5% WB	MCDBR	20.6	20.5	19.0	17.7	17.2	16.5	15.6	15.0	13.8	14.6	17.7	18.7		
		MCWBR	17.9	16.9	14.2	11.7	8.9	8.0	7.9	7.5	7.1	9.5	14.4	16.5		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.332	0.359	0.399	0.444	0.492	0.519	0.502	0.422	0.371	0.342	0.330		
	taud		2.540	2.503	2.446	2.333	2.251	2.162	2.098	2.138	2.369	2.487	2.518	2.542		
	Ebn at Noon		282	290	290	281	269	255	247	249	266	272	271	271		
	Edh at Noon		26	30	34	40	44	48	51	48	36	30	26	24		
(44) All-Sky Solar Radiation	RadAvg		822	1054	1392	1761	1945	2018	1908	1735	1437	1212	912	731		
	RadStd		50	103	102	120	133	109	113	102	96	129	64	55		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.90	N/A	N/A	N/A	N/A	N/A	N/A	+265	+196
(47) Regional (38 neighbors)		+0.76	N/A	N/A	N/A	N/A	N/A	N/A	+288	+203

Nomenclature: See separate page